

3 Theory of Quotients—First Principles of Theory of Numbers

Exercises III

1. Express in its simplest form—

$$\frac{x^2}{x-y} + \frac{y^2}{y-x}.$$

$$\begin{aligned} \frac{x^2}{x-y} + \frac{y^2}{y-x} &= \frac{x^2(y-x)}{(x-y)(y-x)} + \frac{y^2(x-y)}{(x-y)(y-x)} && \left(\frac{a}{b} + \frac{p}{q} = \frac{qa}{qb} + \frac{pb}{qb}\right) \\ &= \frac{x^2(y-x) + y^2(x-y)}{(x-y)(y-x)} && \left(\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}\right) \\ &= \frac{x^2(y-x) + y^2(x-y)}{xy - x^2 - y^2 + xy} && \text{(expand denominator)} \\ &= \frac{x^2(y-x) + y^2(x-y)}{x(y-x) + y(x-y)} && \text{(factor denominator)} \\ &= \frac{x^2(y-x) - y^2(y-x)}{x(y-x) - y(y-x)} && (x-y = -(y-x)) \\ &= \frac{(x^2 - y^2)(y-x)}{(x-y)(y-x)} && \text{(extract common factor } y-x) \\ &= \frac{(x^2 - y^2)}{(x-y)} && \left(\frac{ac}{bc} = \frac{a}{b}\right) \\ &= \frac{(x-y)(x+y)}{(x-y)} && (a^2 - b^2 = (a-b)(a+b)) \\ &= x+y && \left(\frac{ac}{bc} = \frac{a}{b}\right) \end{aligned}$$

2. Express in its simplest form—

$$\frac{a}{a-b} + \frac{b}{b-a}.$$

$$\begin{aligned} \frac{a}{a-b} + \frac{b}{b-a} &= \frac{a(b-a) + b(a-b)}{(a-b)(b-a)} && \left(\frac{a}{b} + \frac{p}{q} = \frac{aq}{bq} + \frac{bp}{bq}\right) \\ &= \frac{a(b-a) + b(a-b)}{ab - a^2 - b^2 + ab} && \text{(expand denominator)} \\ &= \frac{a(b-a) + b(a-b)}{a(b-a) + b(a-b)} && \text{(factor denominator)} \\ &= 1 && \left(\frac{a}{a} = 1\right) \end{aligned}$$

3. Simplify—

$$\frac{P+Q}{P-Q} - \frac{P-Q}{P+Q}, \quad \text{where } P = x+y, \quad Q = x-y.$$

$$\begin{aligned} \frac{P+Q}{P-Q} - \frac{P-Q}{P+Q} &= \frac{x+y+x-y}{x+y-x+y} - \frac{x+y-x+y}{x+y+x-y} && \text{(replace variables)} \\ &= \frac{2x}{2y} - \frac{2y}{2x} && \text{(simplify)} \\ &= \frac{x}{y} - \frac{y}{x} && \left(\frac{ab}{bc} = \frac{a}{c}\right) \\ &= \frac{x^2 - y^2}{xy} && \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \end{aligned}$$

4. Simplify—

$$\frac{1 - \frac{x(1-y)}{x+y}}{1 + \frac{1-y}{x+y}}.$$

$$\begin{aligned}
1 - \frac{x(1-y)}{x+y} \div 1 + \frac{1-y}{x+y} &= \frac{x+y}{x+y} - \frac{x(1-y)}{x+y} \div \frac{x+y}{x+y} + \frac{1-y}{x+y} \\
&= \frac{x+y-x+xy}{x+y} \div \frac{x+y+1-y}{x+y} \\
&= \frac{xy+y}{x+y} \div \frac{x+1}{x+y} \\
&= \frac{xy+y}{x+y} \times \frac{x+y}{x+1} \\
&= \frac{xy+y}{x+1} \\
&= \frac{y(x+1)}{x+1} \\
&= y
\end{aligned}$$

5. Simplify—

$$\frac{\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc}}{\frac{a^2 - (b-c)^2}{a}}$$

$$\begin{aligned}
\frac{\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc}}{\frac{a^2 - (b-c)^2}{a}} &= \left(\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc} \right) \div \frac{a^2 - (b-c)^2}{a} && \left(\frac{a}{b} = a \div b \right) \\
&= \left(\frac{c}{abc} - \frac{b}{abc} - \frac{a}{abc} \right) \div \frac{a^2 - (b-c)^2}{a} && \text{(common denominator } abc) \\
&= \frac{c-b-a}{abc} \div \frac{a^2 - (b-c)^2}{a} && \left(\frac{a}{b} - \frac{c}{b} = \frac{a-c}{b} \right) \\
&= \frac{c-b-a}{abc} \div \frac{(a-(b-c))(a+(b-c))}{a} && (a^2 - b^2 = (a-b)(a+b)) \\
&= \frac{c-b-a}{abc} \div \frac{(a-b+c)(a+b-c)}{a} && \text{(remove brackets)} \\
&= \frac{c-b-a}{abc} \times \frac{a}{(a-b+c)(a+b-c)} && \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p} \right) \\
&= \frac{-(a+b-c)}{abc} \times \frac{a}{(a-b+c)(a+b-c)} && (c-b-a = -(a+b-c)) \\
&= \frac{-a(a+b-c)}{abc(a-b+c)(a+b-c)} && \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= -\frac{1}{bc(a-b+c)} && \left(\frac{ac}{bc} = \frac{a}{b} \right)
\end{aligned}$$

6. Simplify—

$$\left(a - \frac{b^2}{a+b}\right) \times \left(a + \frac{b^2}{a-b}\right).$$

$$\begin{aligned} & \left(a - \frac{b^2}{a+b}\right) \times \left(a + \frac{b^2}{a-b}\right) \\ &= \left(\frac{a(a+b)}{a+b} - \frac{b^2}{a+b}\right) \times \left(\frac{a(a-b)}{a-b} + \frac{b^2}{a-b}\right) \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \\ &= \left(\frac{a^2+ab-b^2}{a+b}\right) \times \left(\frac{a^2-ab+b^2}{a-b}\right) \quad \left(\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}\right) \\ &= \frac{(a^2+ab-b^2)(a^2-ab+b^2)}{(a+b)(a-b)} \quad \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq}\right) \\ &= \frac{a^4 - a^2b^2 + 2ab^3 - b^4}{(a+b)(a-b)} \quad (\text{expand numerator}) \\ &= \frac{a^4 - a^2b^2 + 2ab^3 - b^4}{a^2 - b^2} \quad (\text{distribute denominator}) \end{aligned}$$

7. Simplify—

$$\frac{1}{x+y} + \frac{1}{x-y} - \frac{2x}{x^2+y^2}.$$

$$\begin{aligned} & \frac{1}{x+y} + \frac{1}{x-y} - \frac{2x}{x^2+y^2} \\ &= \frac{x-y+x+y}{(x+y)(x-y)} - \frac{2x}{x^2+y^2} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \\ &= \frac{2x}{x^2-y^2} - \frac{2x}{x^2+y^2} \quad (\text{collect like terms; } (a+b)(a-b) = a^2 - b^2) \\ &= \frac{2x(x^2+y^2) - 2x(x^2-y^2)}{(x^2-y^2)(x^2+y^2)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq-pb}{bq}\right) \\ &= \frac{2x^3 + 2xy^2 - 2x^3 + 2xy^2}{(x^2-y^2)(x^2+y^2)} \quad (\text{expand numerator}) \\ &= \frac{4xy^2}{x^4 + x^2y^2 - y^2x^2 - y^4} \quad (\text{collect like terms; expand denominator}) \\ &= \frac{4xy^2}{x^4 - y^4} \quad (\text{collect like terms}) \end{aligned}$$

8. Simplify—

$$\left(\frac{x}{y} + \frac{y}{x}\right) \left(\frac{a}{b} + \frac{b}{a}\right) - \left(\frac{x}{y} - \frac{y}{x}\right) \left(\frac{a}{b} - \frac{b}{a}\right).$$

$$\begin{aligned}
& \left(\frac{x}{y} + \frac{y}{x}\right) \left(\frac{a}{b} + \frac{b}{a}\right) - \left(\frac{x}{y} - \frac{y}{x}\right) \left(\frac{a}{b} - \frac{b}{a}\right) \\
&= \frac{x^2 + y^2}{xy} \times \frac{a^2 + b^2}{ab} - \frac{x^2 - y^2}{xy} \times \frac{a^2 - b^2}{ab} & \left(\frac{a}{b} \pm \frac{p}{q} = \frac{aq \pm bp}{bq}\right) \\
&= \frac{(x^2 + y^2)(a^2 + b^2)}{abxy} - \frac{(x^2 - y^2)(a^2 - b^2)}{abxy} & \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq}\right) \\
&= \frac{(x^2 + y^2)(a^2 + b^2) - (x^2 - y^2)(a^2 - b^2)}{abxy} & \left(\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b}\right) \\
&= \frac{a^2x^2 + b^2x^2 + a^2y^2 + b^2y^2 - (a^2x^2 - b^2x^2 - a^2y^2 + b^2y^2)}{abxy} & \text{(expand numerator)} \\
&= \frac{a^2x^2 + b^2x^2 + a^2y^2 + b^2y^2 - a^2x^2 + b^2x^2 + a^2y^2 - b^2y^2}{abxy} & \text{(remove brackets)} \\
&= \frac{2b^2x^2 + 2a^2y^2}{abxy} & \text{(collect like terms)} \\
&= \frac{2(b^2x^2 + a^2y^2)}{abxy} & \text{(extract common factor 2)}
\end{aligned}$$

9. Simplify—

$$\left(\frac{a}{b} - \frac{b}{a}\right) \div \left(\frac{a^2}{b^2} - \frac{b^2}{a^2}\right).$$

$$\begin{aligned}
& \left(\frac{a}{b} - \frac{b}{a}\right) \div \left(\frac{a^2}{b^2} - \frac{b^2}{a^2}\right) \\
&= \frac{a^2 - b^2}{ab} \div \frac{a^4 - b^4}{a^2b^2} & \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}\right) \\
&= \frac{a^2 - b^2}{ab} \times \frac{a^2b^2}{a^4 - b^4} & \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p}\right) \\
&= \frac{a^2 - b^2}{ab} \times \frac{a^2b^2}{(a^2 - b^2)(a^2 + b^2)} & (a^2 - b^2 = (a - b)(a + b)) \\
&= \frac{a^2b^2(a^2 - b^2)}{ab(a^2 - b^2)(a^2 + b^2)} & \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq}\right) \\
&= \frac{ab}{a^2 + b^2} & \left(\frac{ac}{bc} = \frac{a}{b}\right)
\end{aligned}$$

10. Simplify—

$$\frac{a(a - b) - b(a + b)}{\frac{a}{a + b} - \frac{b}{a - b}}.$$

$$\begin{aligned}
& \frac{a(a-b) - b(a+b)}{\frac{a}{a+b} - \frac{b}{a-b}} \\
&= (a(a-b) - b(a+b)) \div \left(\frac{a}{a+b} - \frac{b}{a-b} \right) & \left(\frac{a}{b} = a \div b \right) \\
&= (a(a-b) - b(a+b)) \div \frac{a(a-b) - b(a+b)}{(a+b)(a-b)} & \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= (a(a-b) - b(a+b)) \times \frac{(a+b)(a-b)}{a(a-b) - b(a+b)} & \left(a \div \frac{p}{q} = a \times \frac{q}{p} \right) \\
&= (a+b)(a-b) & \left(\frac{ac}{bc} = \frac{a}{b} \right) \\
&= a^2 - b^2 & ((a+b)(a-b) = a^2 - b^2)
\end{aligned}$$

11. Simplify—

$$\frac{1-x}{1+x+x^2} - \frac{1+x}{1-x+x^2}.$$

$$\begin{aligned}
& \frac{1-x}{1+x+x^2} - \frac{1+x}{1-x+x^2} \\
&= \frac{(1-x)(1-x+x^2) - (1+x)(1+x+x^2)}{(1+x+x^2)(1-x+x^2)} & \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= \frac{1-2x+2x^2-x^3 - (1+2x+2x^2+x^3)}{(1+x+x^2)(1-x+x^2)} & \text{(expand numerator)} \\
&= \frac{-4x-2x^3}{(1+x^2)^2 - x^2} & ((p+q)(p-q) = p^2 - q^2) \\
&= \frac{-4x-2x^3}{1+x^2+x^4} & \text{(expand denominator)}
\end{aligned}$$

12. Simplify—

$$\frac{1 + \frac{x}{1+x}}{x + \frac{1}{1+x}} \div \frac{(x+1)^2 - x^2}{x^2 + x + 1}.$$

$$\begin{aligned}
& \frac{1 + \frac{x}{1+x}}{x + \frac{1}{1+x}} \div \frac{(x+1)^2 - x^2}{x^2 + x + 1} \\
&= \frac{\frac{1+2x}{1+x}}{\frac{x^2+x+1}{1+x}} \div \frac{2x+1}{x^2+x+1} \quad (\text{common denominator } 1+x; \text{ expand}) \\
&= \frac{1+2x}{1+x} \times \frac{1+x}{x^2+x+1} \div \frac{2x+1}{x^2+x+1} \quad \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p}\right) \\
&= \frac{1+2x}{x^2+x+1} \times \frac{x^2+x+1}{2x+1} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right) \\
&= 1 \quad \left(\frac{a}{a} = 1\right)
\end{aligned}$$

13. Simplify—

$$\frac{a^2 + b^2}{(a+b)^2} + \frac{\frac{2}{ab} \left(\frac{1}{a} + \frac{1}{b}\right)}{\left(\frac{1}{a} + \frac{1}{b}\right)^3}.$$

$$\begin{aligned}
& \frac{a^2 + b^2}{(a+b)^2} + \frac{\frac{2}{ab} \left(\frac{1}{a} + \frac{1}{b}\right)}{\left(\frac{1}{a} + \frac{1}{b}\right)^3} \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{1}{\left(\frac{1}{a} + \frac{1}{b}\right)^2} \quad \left(\frac{ac}{bc} = \frac{a}{b}, \text{ cancelling } \frac{1}{a} + \frac{1}{b}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{1}{\left(\frac{a+b}{ab}\right)^2} \quad \left(\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{a^2 b^2}{(a+b)^2} \quad \left(\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2ab}{(a+b)^2} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right) \\
&= \frac{a^2 + 2ab + b^2}{(a+b)^2} \quad \left(\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}\right) \\
&= \frac{(a+b)^2}{(a+b)^2} \quad (a^2 + 2ab + b^2 = (a+b)^2) \\
&= 1 \quad \left(\frac{a}{a} = 1\right)
\end{aligned}$$

14. Show that—

$$\frac{x^4}{a^2b^2} + \frac{(x^2 - a^2)^2}{a^2(a^2 - b^2)} - \frac{(x^2 - b^2)^2}{b^2(a^2 - b^2)}$$

is independent of x .

Proof. Taking $a^2b^2(a^2 - b^2)$ as a common denominator,

$$\frac{x^4}{a^2b^2} + \frac{(x^2 - a^2)^2}{a^2(a^2 - b^2)} - \frac{(x^2 - b^2)^2}{b^2(a^2 - b^2)} = \frac{N}{a^2b^2(a^2 - b^2)},$$

where

$$\begin{aligned} N &= x^4(a^2 - b^2) + b^2(x^2 - a^2)^2 - a^2(x^2 - b^2)^2 \\ &= x^4(a^2 - b^2) + b^2(x^4 - 2a^2x^2 + a^4) \\ &\quad - a^2(x^4 - 2b^2x^2 + b^4) && ((p - q)^2 = p^2 - 2pq + q^2) \\ &= a^2x^4 - b^2x^4 + b^2x^4 - 2a^2b^2x^2 + a^4b^2 \\ &\quad - a^2x^4 + 2a^2b^2x^2 - a^2b^4 && (\text{remove brackets}) \\ &= a^4b^2 - a^2b^4 && (\text{collect like terms}) \\ &= a^2b^2(a^2 - b^2) && (\text{extract common factor } a^2b^2) \end{aligned}$$

Then the expression is

$$\frac{a^2b^2(a^2 - b^2)}{a^2b^2(a^2 - b^2)} = 1.$$

From this we see that all terms of x cancel, so its value is always 1, thus the expression is independent of x . $\square$

15. Simplify—

$$\frac{a}{b - \frac{c}{d - \frac{e}{f}}}.$$

$$\begin{aligned}
& \frac{a}{b - \frac{c}{d - \frac{e}{f}}} \\
&= a \div \left[b - \left(c \div \left(d - \frac{e}{f} \right) \right) \right] & \left(\frac{a}{b} = a \div b \right) \\
&= a \div \left[b - \left(c \div \frac{df - e}{f} \right) \right] & \left(a - \frac{p}{q} = \frac{aq - p}{q} \right) \\
&= a \div \left[b - \frac{cf}{df - e} \right] & \left(a \div \frac{p}{q} = \frac{aq}{p} \right) \\
&= a \div \left[\frac{b(df - e) - cf}{df - e} \right] & \left(a - \frac{p}{q} = \frac{aq - p}{q} \right) \\
&= \frac{a(df - e)}{b(df - e) - cf} & \left(a \div \frac{p}{q} = \frac{aq}{p} \right) \\
&= \frac{adf - ae}{bdf - be - cf} & (\text{expand})
\end{aligned}$$

16. Simplify—

$$\frac{1}{a - 2b - \frac{1}{a - 2b - \frac{1}{a - 2b}}}$$

Writing $t = a - 2b$,

$$\begin{aligned}
& \frac{1}{a - 2b - \frac{1}{a - 2b - \frac{1}{a - 2b}}} \\
&= \frac{1}{t - \frac{1}{t - \frac{1}{t}}} && \text{(replace variables)} \\
&= \frac{1}{t - \frac{1}{\frac{t^2 - 1}{t}}} && (\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}) \\
&= \frac{1}{t - \frac{t}{t^2 - 1}} && (a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{1}{\frac{t(t^2 - 1) - t}{t^2 - 1}} && (\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}) \\
&= \frac{t^2 - 1}{t^3 - 2t} && \text{(expand; } a \div \frac{p}{q} = \frac{aq}{p} \text{)} \\
&= \frac{(a - 2b)^2 - 1}{(a - 2b)^3 - 2(a - 2b)} && \text{(restore variables)}
\end{aligned}$$

17. Simplify—

$$\frac{a + b}{a + b + \frac{1}{a - b + \frac{1}{a + b}}}.$$

$$\begin{aligned}
& \frac{a+b}{a+b+\frac{1}{a-b+\frac{1}{a+b}}} \\
&= \frac{a+b}{a+b+\frac{1}{(a-b)(a+b)+1}} \quad \left(\frac{a}{b} + \frac{p}{q} = \frac{aq+pb}{bq}\right) \\
&= \frac{a+b}{a+b+\frac{a+b}{a^2-b^2+1}} \quad ((a+b)(a-b) = a^2-b^2; a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{a+b}{\frac{(a+b)(a^2-b^2+1) + (a+b)}{a^2-b^2+1}} \quad \left(\frac{a}{b} + \frac{p}{q} = \frac{aq+pb}{bq}\right) \\
&= \frac{a+b}{\frac{(a+b)(a^2-b^2+2)}{a^2-b^2+1}} \quad (\text{extract common factor } a+b) \\
&= \frac{(a+b)(a^2-b^2+1)}{(a+b)(a^2-b^2+2)} \quad \left(a \div \frac{p}{q} = \frac{aq}{p}\right) \\
&= \frac{a^2-b^2+1}{a^2-b^2+2} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right)
\end{aligned}$$

Exercises IV

1. If the two fractions A/B , a/b be equal, and the latter be at its lowest terms, prove that $A = \mu a$, $B = \mu b$, where μ is an integer.

Proof. Suppose A , B , a , b integers with $A/B = a/b$, and a/b at its lowest terms; that is, a and b have no common measure save unity, so that a is prime to b (§6). Multiplying across by bB ,

$$Ab = aB;$$

hence a divides Ab , and being prime to b it must divide A (§10, Cor. 1). Write $A = \mu a$, μ an integer. Substituting, $\mu ab = aB$, and cancelling a (which is not zero), $B = \mu b$ —the same μ serving in both. $\square$

2. Prove that the sum or difference of two odd numbers is always even; the sum or difference of an odd and an even number always odd; the product of any number of odd numbers always odd; the quotient of one odd number by another always odd, if it be integral.

Proof. Let a, b, c be integers where a is even and b and c are odd. Then $a = 2m$, $b = 2n + 1$, and $c = 2l + 1$ for some integers m, n, l (Cor., §12). Next, we prove the propositions individually.

1. *Sum or difference of two odd numbers is always even.* Take $b + c$, the sum of two odd numbers, this can be written $b + c = (2n + 1) + (2l + 1) = 2n + 1 + 2l + 1 = 2n + 2l + 2 = 2(n + l + 1)$. Thus, the sum is always even. Now take the difference of two odd numbers, in either order: $b - c = (2n + 1) - (2l + 1) = 2n + 1 - 2l - 1 = 2n - 2l = 2(n - l)$, and $c - b = (2l + 1) - (2n + 1) = 2l + 1 - 2n - 1 = 2l - 2n = 2(l - n)$. Thus the difference is always even.
2. *Sum or difference of an odd and an even number is always odd.* Take $a + b$, the sum of an even number and an odd number, this can be written $a + b = 2m + (2n + 1) = 2(m + n) + 1$. Thus, the sum is always odd. Now take the difference, in either order: $a - b = 2m - (2n + 1) = 2(m - n - 1) + 1$, and $b - a = (2n + 1) - 2m = 2(n - m) + 1$. Thus the difference is always odd.
3. *The product of any number of odd numbers is always odd.* First take two of them, $bc = (2n + 1)(2l + 1) = 4nl + 2n + 2l + 1 = 2(2nl + n + l) + 1$, which is odd. Now suppose the product of k odd numbers to be odd, say $2p + 1$, and let $2q + 1$ be a further odd number. Then $(2p + 1)(2q + 1) = 2(2pq + p + q) + 1$, which is odd again. Hence the product of $k + 1$ odd numbers is odd whenever the product of k of them is; and since the product of two is odd, the product of any number of odd numbers is odd.
4. *The quotient of one odd number by another is always odd, if it be integral.* Let b/c be integral, say $b = cq$, where q is an integer. Were q even, $q = 2k$ say, we should have $b = 2ck$, that is, b even, contrary to hypothesis. Hence q is odd.

□

3. If a be prime to b , then—

- a) $(a + b)^m$ and $(a - b)^m$ have at most the G.C.M. 2^m ;
- b) $a^m + b^m$ and $a^m - b^m$ have at most the G.C.M. 2 ;
- c) $a + b$ and $a^2 + b^2 - ab$ have at most the G.C.M. 3 .

Lemma A. If x be prime to y , and g divide $x + y$, then g is prime to x and to y .

Proof. Let e be a common measure of g and x . Then e divides $x + y$ and x . Therefore e divides $(x + y) - x = y$ (§8). Thus e is a common measure of x and y . Because x and y are prime to each other, it follows e is unity. Hence g is prime to x , and in like manner g is prime to y . □

Lemma B. If x be prime to y , the G.C.M. of $x + y$ and $x - y$ divides 2.

Proof. Let g be the G.C.M. of $x + y$ and $x - y$. Then g divides $(x + y) + (x - y) = 2x$ (§8). Since g is prime to x (Lemma A), it follows g divides 2 (§10, Cor.1). $\square$

Proof of (a). Let g be the G.C.M. of $a + b$ and $a - b$. Then g divides 2 (Lemma B). Next, write $a + b = gp$ and $a - b = gq$ for integers p, q . Then p is prime to q ; for if they had a common factor e greater than unity, ge would be a common factor of $a + b$ and $a - b$ greater than their G.C.M. Hence p^m is prime to q^m (§10, Cor. 4), and integers A, B can be found such that $Ap^m - Bq^m = \pm 1$ (§14, Cor. 2). Thus $Ag^mp^m - Bg^mq^m = \pm g^m$. Now $(a + b)^m = g^mp^m$ and $(a - b)^m = g^mq^m$, so that every common factor of $(a + b)^m$ and $(a - b)^m$ divides this algebraic sum of their multiples (§8), that is, divides g^m . But g divides 2, and therefore g^m divides 2^m ; hence the G.C.M. of $(a + b)^m$ and $(a - b)^m$ is at most 2^m . $\square$

Proof of (b). Since a is prime to b , a^m is prime to b^m (§10, Cor. 4). Hence, by Lemma B applied to a^m and b^m , the G.C.M. of $a^m + b^m$ and $a^m - b^m$ divides 2, that is, is at most 2. $\square$

Proof of (c). Let g be the G.C.M. of $a + b$ and $a^2 + b^2 - ab$. Since g divides $a + b$ it divides $(a + b)^2$, and therefore divides $(a + b)^2 - (a^2 + b^2 - ab) = 3ab$ (§8). Since g divides $a + b$, it follows g is prime to a and b (Lemma A). Hence g is prime to their product ab (§10, Cor. 2). But g divides $3ab$ and is prime to ab , so g divides 3 (§10, Cor. 1), that is, the G.C.M. of $a + b$ and $a^2 + b^2 - ab$ is at most 3. $\square$

4. The difference of the squares of any two odd numbers is exactly divisible by 8.

Proof. Let a and b be odd, so that $a = 2k + 1$ and $b = 2l + 1$ for some integers k, l . Then

$$a^2 - b^2 = (4k^2 + 4k + 1) - (4l^2 + 4l + 1) = 4[(k^2 + k) - (l^2 + l)].$$

Now, k is either even or odd; if k is odd then k^2 is odd (Exercise 2, part 3) thus $k^2 + k$ is even (Exercise 2, part 1); if k is even, then k^2 and k are both divisible by 2, and so is their sum (§8), hence $k^2 + k$ is even. So we have in either case $k^2 + k$ is even. In likely manner $l^2 + l$ is even. That is, $k^2 + k = 2m$ and $l^2 + l = 2n$ for some integers m, n ; whence

$$a^2 - b^2 = 4(2m - 2n) = 8(m - n).$$

Therefore $a^2 - b^2$ is exactly divisible by 8. $\square$

5. The sum of the squares of three consecutive odd numbers increased by 1 is a multiple of 12.

Proof. Let a, b, c be three consecutive odd numbers, so that $a = 2k - 1$, $b = 2k + 1$ and $c = 2k + 3$ for some integer k . Then

$$\begin{aligned} a^2 + b^2 + c^2 + 1 &= (2k - 1)^2 + (2k + 1)^2 + (2k + 3)^2 + 1 \\ &= (4k^2 - 4k + 1) + (4k^2 + 4k + 1) + (4k^2 + 12k + 9) + 1 \\ &= 12k^2 + 12k + 12 \\ &= 12(k^2 + k + 1) \end{aligned}$$

Therefore $a^2 + b^2 + c^2 + 1$ is a multiple of 12. □

6. If each of two fractions be at its lowest terms, neither their sum nor their difference can be an integer unless the denominators be equal.

Proof. Let a/b and c/d be the fractions both at its lowest terms; that is, a is prime to b and c is prime to d . Suppose their sum to be an integer n so that

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm cb}{bd} = n$$

Hence $ad \pm cb = nbd$. Then $ad = nbd \mp cb = b(nd \mp c)$, so b divides ad , since a and b are prime to each other, we have that b divides d . Similarly $cb = nbd \mp ad = d(nb \mp a)$ so d divides cb , since d is prime to c , we have that d divides b . Because d divides b and b divides d they must be equal, that is $b = d$. □

7. Resolve 45738 and 297675 into their prime factors.

$$\begin{aligned} 45738 &= 2 \times 3^3 \times 7 \times 11^2 \\ 297675 &= 3^5 \times 5^2 \times 7^2 \end{aligned}$$

8. Find the G.C.M. of 54643 and 91319, using negative remainders whenever it is of advantage to do so.

9. Prove that the L.C.M. of two integers is the quotient of their product by the G.C.M.

10. If g_1, g_2, g_3 be the G.C.M.'s, l_1, l_2, l_3 the L.C.M.'s, of b and c , c and a , a and b respectively, G the G.C.M., and L the L.C.M., of the three a, b, c , show that

$$\text{a) } L = \frac{abcG}{g_1g_2g_3};$$

$$\text{b) } \frac{L}{G} = \sqrt{\frac{l_1 l_2 l_3}{g_1 g_2 g_3}}.$$

11. When x is divided by y , the quotient is u and the remainder v ; show that, when x and uy are divided by v , the remainders are the same, and the quotients differ by unity.